

Case 2: $x \geq 0, y < 0$.

$$\therefore (-1) \cdot y > -1 \cdot 0 \Rightarrow (-y) > 0.$$

$$\therefore x + (-y) > x + 0 \Rightarrow \boxed{x + (-y) > x}$$

$$y < 0 \Rightarrow x + y < 0 + x \Rightarrow x + y < x$$

If $(x+y) > 0$, $|x+y| = x+y$ (By defn.) $\therefore |x+y| < x < |x| + |y|$ (By defn.)
 $\therefore |x| + |y| > |x+y|$ holds when $(x+y) > 0$

If $(x+y) < 0$, $|x+y| = -(x+y)$ (By defn.) $= -x - y$

$$\therefore x \geq 0 \Rightarrow (-1) \cdot x < (-1) \cdot 0 \Rightarrow -x < 0 \Rightarrow -x - y < 0 - y \Rightarrow -x - y < -y$$

$$x - y > 0 - y \Rightarrow x - y > -y \quad \therefore x - y > -x - y \Rightarrow x + (-y) > -(x+y)$$

Case 3: $x < 0, y \geq 0$ $\therefore$ By defn, $|x| = -x$ and $|y| = y$
 $|x| + |y| = -x + y$. If $(x+y) > 0$, then $|x+y| = x+y$

$$(-1) \cdot x > (-1) \cdot 0 \Rightarrow -x > 0 \Rightarrow -x + y > y$$

$$x + y < 0 + y \Rightarrow x + y < y$$

$$\therefore \boxed{x + y > y} \Rightarrow \boxed{|x| + |y| > |x+y|}$$

If $(x+y) < 0$, then $|x+y| = -(x+y) = -x - y$

$$x + y < 0 \Rightarrow -x - y > 0 + (-x) \Rightarrow -x - y > -x$$

$$-y < 0 \Rightarrow -x - y < 0 - y \Rightarrow -x - y < -y \quad \therefore -x - y > -x - y \Rightarrow \boxed{|x| + |y| > |x+y|}$$

Case 4: $x < 0, y < 0$ $\therefore$ By defn, $|x| = -x$, $|y| = -y$
 $\therefore x + y < 0$ $\therefore$ By defn, $|x+y| = -(x+y)$

$$|x| + |y| = (-x) + (-y) = -(x+y) = |x+y| \quad \therefore \boxed{|x| + |y| \geq |x+y|} \quad \square$$